An Improved Ship Detection Algorithm for an Airborne Passive Interferometric Microwave Sensor (PIMS) Based on Ship Wakes

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***Abstract*— Airborne passive interferometric microwave sensors (PIMSs) were proposed as a powerful complementary tool for ship detection, especially in low visibility and high sea clutter environments. At present, there is only one algorithm for ship detection by airborne PIMSs, and this algorithm is only based on the "cold" brightness temperature (TB) characteristic of metal ships with wakes in passive interferometric microwave images (PIMIs). Moreover, it has been found that ship wakes always exhibit a "hot" TB characteristic in PIMIs. In this article, an improved algorithm is innovatively proposed for ship detection by airborne PIMSs to improve detection performance, and it is simultaneously based on the "cold" and "hot" TB characteristics of metal ships with wakes. The microwave TB model of metal ships with wakes is established to analyze its TB characteristics. Given the TB model and its characteristics, an improved algorithm is introduced, which comprises five steps: searching targets, improving TB maps, clustering targets, estimating ship headings, and estimating ship trajectories. The improved algorithm can detect ships with a low false-alarm probability (FAR) and accurately estimate ship headings using only a single snapshot and the ship trajectory. The practicability and detection performance of the improved algorithm is verified by airborne experiments using an X-band PIMS. The experiment results demonstrate that the improved algorithm can accurately detect ships, estimate ship headings using only a single snapshot, and estimate ship trajectories. The impact of the algorithm parameters on the detection performance is analyzed and discussed. Moreover, the detection performance of the improved algorithm is analyzed and compared with the performance of the only existing algorithm, and the experiment results indicate that the improved algorithm has a better detection performance with a lower FAR value than the only existing algorithm, as expected.**

***Index Terms*— Airborne, improved ship detection, interferometric, microwave sensor, passive, ship wakes.**

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NOMENCLATURE

AIS	Automatic identification system.
SAR	Synthetic aperture radar.
PIMS	Passive interferometric microwave sensor.
FOV	Field of view.
TB	Brightness temperature.
PIMI	Passive interferometric microwave image.
FAR	False-alarm rate.
FFT	Fast Fourier transform.

I. INTRODUCTION

SHIP detection from airborne/spaceborne platforms is a hot topic in the field of remote sensing and is of vital importance due to its wide range of applications, e.g., oceanic traffic surveillance and management, ship rescue, oil-spill detection, fishery management, and illegal smuggling [1], [2]. Currently, ships on the ocean can be detected by several airborne/spaceborne sensors, e.g., AISs [3], [4], optical sensors [5], infrared sensors [6], [7], and SAR [8], [9]. However, these sensors have their advantages and weaknesses, especially in low visibility conditions and/or high sea clutter environments [10], [11], [12]. For example, optical sensors and infrared sensors usually suffer from interference by low visibility conditions, and SAR usually suffers from sea clutter and high-level speckles [10], [11]. Fortunately, passive microwave imaging has efficient imaging capability in low visibility conditions, has no speckle noise issues, and slightly suffers from sea clutter [10]. Moreover, PIMSs have been proposed to realize high-resolution passive microwave imaging with a wide FOV and a high frame rate [13], [14], [15]. Lu et al. [16] proposed and demonstrated that ships can be accurately detected and consecutively tracked by airborne PIMSs, which is considered a powerful complementary tool for ship detection, especially in low visibility and high sea clutter environments. These sensors can complement each other for ship detection during all times of day and in all weather conditions [16].

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For an airborne PIMS, ship detection is based on the high TB contrast of ships in PIMIs [16]. However, since the systematic noises, systematic errors, and fluctuations of the aircraft have greatly reduced the TB contrast of the ships in PIMIs, ships cannot be detected by the airborne PIMSs in raw observed TB maps [17]. Thus, Lu et al. [17] proposed a ship detection algorithm for airborne PIMSs, which is only based on the "cold" TB characteristic of ships in PIMIs. Furthermore, it is interesting to find that ship wakes always exhibit a "hot" TB characteristic in a PIMI, which can be considered another important characteristic for ship detection by airborne PIMSs, as mentioned in [16] and [18]. At present, the ship detection algorithm proposed by Lu et al. [17] is the only existing ship detection algorithm for airborne PIMSs. To further improve the detection performance, an improved ship detection algorithm is proposed in this article, which is simultaneously based on the "cold" TB characteristic of metal ships and the "hot" TB characteristic of ship wakes in PIMIs. The improved ship detection algorithm can detect ships with a high detection probability and a low FAR compared with the only existing ship detection algorithm [17]. In addition, the ship heading can be accurately estimated using only a single snapshot by the improved algorithm.

The remainder of this article is organized as follows. In Section II, the TB model of metal ships with ship wakes is established, and its characteristics are analyzed. In Section III, the improved ship detection algorithm is introduced. Then, the improved algorithm is verified by airborne experiments, and the results and discussion are given in Section IV. Finally, the conclusion is drawn.

II. TB Models and Characteristics of Metal Ships With Wakes

It is well-known that the observed TB map of the sea background by an airborne PIMS can be expressed in direction cosines (ξ, η) as

$$\mathrm{TB}_{\mathrm{map}}^{\mathrm{sea}}(\xi,\eta) = \mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}(\xi,\eta) + \Delta T_{\mathrm{sys}}(\xi,\eta) + T_n(\xi,\eta) \quad (1)$$

where the direction cosines $(\xi, \eta) = (\sin\theta\cos\phi, \sin\theta\sin\phi)$, $\mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}$ is the apparent TB of the sea surface, ΔT_{sys} denotes the residual additional systematic errors or the additional systematic errors, which are considered to be additive, and are almost the same in all frames [19], and T_n denotes the systematic noise in PIMIs [17]. In addition, the adjacent apparent TB of the sea surface $\mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}$ from the air is considered to be almost the same in an open ocean [20], [21].

For an airborne PIMS, the apparent TB of a ship can be expressed as

$$\mathrm{TB}_{\mathrm{Ap}}^{\mathrm{ship}} = e^{\mathrm{ship}} \cdot T^{\mathrm{ship}} \cdot \tau_{\mathrm{Up}} + T_{\mathrm{DN}} \cdot \left(1 - e^{\mathrm{ship}}\right) \cdot \tau_{\mathrm{Up}} + T_{UP} \quad (2)$$

where e^{ship} denotes the microwave emissivity of the ship, T^{ship} denotes the physical temperature of the ship, τ_{UP} denotes the transmissivity between the sea surface and the airborne PIMSs, T_{UP} denotes the upwelling atmospheric radiation between the sea surface and the airborne PIMSs, and T_{DN} denotes the downwelling atmospheric radiation. The emissivity of the ship depends on its materials. The emissivity of a painted metallic ship is usually lower than 0.1 below 100 GHz, which is lower than the emissivity of the ocean surface by approximately 0.4–0.7 [10]. Large ships, such as oil tankers and container ships, are usually made of metals. As a result, metallic ships exhibit a "low" TB characteristic against the sea surface background. However, some small ships made of wooden materials, fiberglass materials, and so on, exhibit a higher emissivity than the ocean surface emissivity.

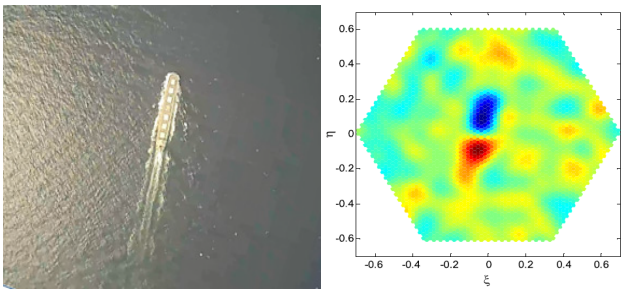

Fig. 1. (Left) Optical image of a ship sailing on the sea. (Right) TB map of the ship sailing on the sea.

For an airborne PIMS, the apparent TB of ship wakes can be expressed as

$$\mathrm{TB}_{Ap}^{\mathrm{wake}} = e^{\mathrm{wake}} \cdot T^{\mathrm{wake}} \cdot \tau_{\mathrm{Up}} + T_{\mathrm{DN}} \cdot \left(1 - e^{\mathrm{wake}}\right) \cdot \tau_{\mathrm{Up}} + T_{\mathrm{UP}} \quad (3)$$

where e^{wake} denotes the emissivity of ship wakes and T^{wake} denotes the physical temperature of ship wakes. The surface roughness in ship wakes is larger than that in the sea surface background, and it contains a large amount of foam, especially at the end of a sailing ship (see Fig. 1). Sea surface foam has a higher emissivity (approximately 1) than the sea background [22]. These characteristics cause ship wakes to exhibit a higher emissivity than the sea background. As a result, ship wakes exhibit a higher TB than the sea background.

As mentioned in [10] and [17], a ship made of painted metals exhibits a high contrast and a "cold" TB characteristic against the sea background in a PIMI, and ship wakes exhibit a high contrast with a "hot" TB characteristic against the sea background in a PIMI. The "hot" TB of the ship wakes is mainly generated by the increased sea surface roughness and foam-covered sea surface caused by the sailing ship. For a sailing metal ship, the "cold" TB characteristic caused by the ship-self is always accompanied by a "hot" TB characteristic caused by the ship wakes. Thus, when the ship wakes are considered, the TB model of a sailing metal ship with wakes can be expressed as

$$\begin{aligned}
&\mathrm{TB}_{\mathrm{map}}^{\mathrm{ship}}(\xi,\eta,n) \\
&= \mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}(\xi,\eta,n) + \Delta T_{\mathrm{sys}}(\xi,\eta,n) + T_n(\xi,\eta,n) \\
&\quad + \left(\mathrm{TB}_{\mathrm{AP}}^{\mathrm{ship}}(\xi,\eta,n) - \mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}(\xi,\eta,n)\right) \\
&\quad \cdot \delta\left(\xi - \xi_0^{\mathrm{ship}}, \eta - \eta_0^{\mathrm{ship}}\right) + \left(\mathrm{TB}_{\mathrm{AP}}^{\mathrm{wake}}(\xi,\eta,n) - \mathrm{TB}_{\mathrm{AP}}^{\mathrm{sea}}(\xi,\eta,n)\right) \\
&\quad \cdot \delta\left(\xi - \xi_0^{\mathrm{wake}}, \eta - \eta_0^{\mathrm{wake}}\right) \quad (4)
\end{aligned}$$

where n is the frame number of the raw observed PIMIs, $(\xi_0^{\mathrm{ship}}, \eta_0^{\mathrm{ship}})$ denotes a pixel of ship "0" on the sea in the PIMI,

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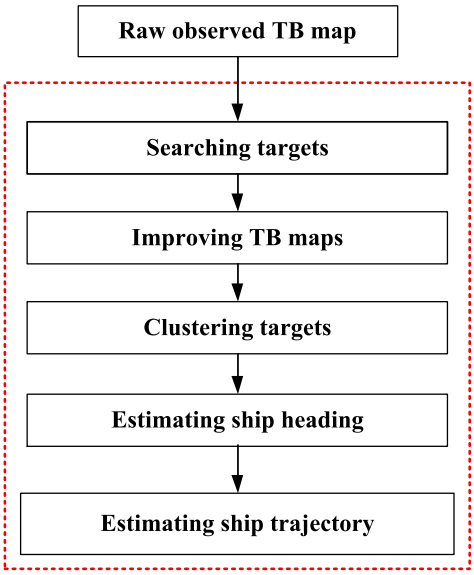

Fig. 2. Flowchart of the improved ship detection algorithm.

$(\xi_0^{\text{wake}}, \eta_0^{\text{wake}})$ denotes a pixel of the wake of ship "0" on the sea in the PIMI, and δ is the impulse function. Since the ship wakes are caused by the sailing ships, the "hot" TB of the ship wakes is always accompanied by the "cold" TB of the ship itself, which can be expressed as

$$\begin{aligned} |\xi_0^{\text{ship}} - \xi_0^{\text{wake}}| &\le \xi_{\min} \\ |\eta_0^{\text{ship}} - \eta_0^{\text{wake}}| &\le \eta_{\min} \end{aligned} \tag{5}$$

in which $\xi_{\min}$ and $\eta_{\min}$ depend on the length and width of the ship and the spatial resolution of the airborne PIMSs.

In addition, the ship heading can be estimated by

$$\vec{H} = \left(\xi_0^{\text{ship}} - \xi_0^{\text{wake}}, \eta_0^{\text{ship}} - \eta_0^{\text{wake}}\right). \tag{6}$$

As shown in Fig. 1, ship wakes weaken over time in terms of the sea surface roughness and foam-covered sea surface, and it can be seen that ship wakes close to the ship exhibit a large surface roughness and contain more foam than ship wakes away from the ship. Thus, ship wakes close to the ship exhibit a higher TB than ship wakes away from the ship.

In addition, it must be noted that the sailing ships are line targets or face targets, and the TB model of the sailing ships in (4) can be established such that $(\xi_0^{\text{ship}}, \eta_0^{\text{ship}})$ and $(\xi_0^{\text{wake}}, \eta_0^{\text{wake}})$ are pixel sets.

III. Improved Ship Detection Algorithm for an Airborne PIMS Based on Ship Wakes

This section is devoted to an improved ship detection algorithm based on ship wakes. The improved ship detection algorithm consists of five steps, and it can not only detect ships on the sea but also estimate the ship heading by a single snapshot. The five steps consist of searching targets, improving maps, clustering targets, estimating ship headings, and estimating ship trajectories, as shown in Fig. 2. The improved ship detection algorithm is introduced in detail as follows.

A. Step 1: Searching Targets

As mentioned in [17], the residual systematic errors and systematic noise weaken or merge the high contrast of the ships in the raw observed maps. This step aims to flag the raw observed TB maps with ships and determine the frame range of the ship in the observed TB maps.

In this step, all raw observed maps are processed to obtain the difference-detection TB maps by subtracting them from their adjacent raw observed maps. If there is a ship in a raw observed map, the difference-detection TB map can be expressed as

$$\begin{aligned} &\Delta T_D(\xi, \eta, n) \\ &= \text{TB}_{\text{map}}^{\text{ship}}(\xi, \eta, n) - \text{TB}_{\text{map}}^{\text{ship}}(\xi, \eta, m) \\ &= \text{TB}_{\text{AP}}^{\text{ship}}\left(\xi_n^{\text{ship}}, \eta_n^{\text{ship}}, n\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_n^{\text{ship}}, \eta_n^{\text{ship}}, n\right) \\ &\quad + \text{TB}_{\text{AP}}^{\text{wake}}\left(\xi_n^{\text{wake}}, \eta_n^{\text{wake}}, n\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_n^{\text{wake}}, \eta_n^{\text{wake}}, n\right) \\ &\quad - \left(\text{TB}_{\text{AP}}^{\text{ship}}\left(\xi_m^{\text{ship}}, \eta_m^{\text{ship}}, m\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_m^{\text{ship}}, \eta_m^{\text{ship}}, m\right)\right) \\ &\quad - \left(\text{TB}_{\text{AP}}^{\text{wake}}\left(\xi_m^{\text{wake}}, \eta_m^{\text{wake}}, m\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_m^{\text{wake}}, \eta_m^{\text{wake}}, m\right)\right) \\ &\quad + \sqrt{2} T_n(\xi, \eta) \end{aligned} \tag{7}$$

where n and m are the frame numbers of the raw observed TB map and $\text{l} = |n - m|$ mainly depends on the fluctuations of the aircraft and can be determined by analyzing the TB fluctuations of one element in the PIMSs [17].

Then, the raw observed maps are flagged as

$$f(n) = \begin{cases} 1, & \text{if } \max\{|\Delta T_D(\xi, \eta, n)|\} \ge N \cdot \sigma_0 \\ 0, & \text{else} \end{cases} \tag{8}$$

in which "1" and "0" denote that there are ships and no ships in the nth observed TB maps, respectively. N is a scaling threshold factor and σ_0 is the standard deviation of the observed TB maps. Thus, the consecutive observed TB maps $(t_1, \ldots, t_s)$ that contain ships are flagged as "1" in the step.

B. Step 2: Improving TB Maps

Since the residual systematic errors and the systematic noise weaken or merge the high contrast of ships in raw observed maps, the ships cannot be clearly seen or identified in raw observed maps. This step aims to improve the contrast of the ships in raw observed maps by subtracting consecutive raw observed TB maps that contain ships from a raw observed TB map of the sea background, and the improved observed TB map is obtained by

$$\begin{aligned} &\Delta T_B(\xi, \eta, n) \\ &= \text{TB}_{\text{map}}^{\text{ship}}(\xi, \eta, n) - \text{TB}_{\text{map}}^{\text{sea}}(\xi, \eta, k) \\ &= \text{TB}_{\text{AP}}^{\text{ship}}\left(\xi_n^{\text{ship}}, \eta_n^{\text{ship}}, n\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_n^{\text{ship}}, \eta_n^{\text{ship}}, n\right) \\ &\quad + \text{TB}_{\text{AP}}^{\text{wake}}\left(\xi_n^{\text{wake}}, \eta_n^{\text{wake}}, n\right) - \text{TB}_{\text{AP}}^{\text{sea}}\left(\xi_n^{\text{wake}}, \eta_n^{\text{wake}}, n\right) \\ &\quad + \sqrt{2} T_n(\xi, \eta) \end{aligned} \tag{9}$$

in which the raw kth observed map is the sea background with no ship. In the improved observed TB maps, a ship exhibits a "cold" TB characteristic with a high contrast, and the ship wake exhibits a "hot" TB characteristic with a high contrast.

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In addition, as mentioned in [17], since aircraft fluctuations produce additional errors that reduce the "hot" and "cold" TB characteristics in the improved observed TB maps, the raw observed TB map of the sea background should be carefully selected in this step to reduce the negative impact of aircraft fluctuations. By analyzing the TB fluctuations of one element in the PIMSs, the "ship-free" observed TB maps are carefully selected to represent the sea background in (9), and the sea background in these images should be the same as the sea background in the observed TB maps that contain ships.

C. Step 3: Clustering Targets

Based on the TB characteristic of the ship and the ship wake, the first target flag in the improved observed TB maps is determined as follows:

$$F_1(\xi,\eta,n)=\begin{cases}1, & \Delta T_B(\xi,\eta,n)\le -M_1\cdot\sigma_0\\ -1, & \Delta T_B(\xi,\eta,n)\ge M_2\cdot\sigma_0\\ 0, & \text{else}\end{cases}\tag{10}$$

where "1" denotes that the pixel contains a ship (called "ship-containing" pixels), "−1" denotes that the pixel contains a ship wake (called "ship wake-containing" pixels), and "0" denotes that the pixel contains no ships or ship wakes. $M1$ and $M2$ are also scaling threshold factors. Then, the target cluster [23] is found such that $\xi_{s_i}^{\text{ship}}\in\xi_s^{\text{ship}}(i=1,\ldots,l_1)$, $\eta_{s_i}^{\text{ship}}\in\eta_s^{\text{ship}}(i=1,\ldots,l_1)$, $\xi_{w_j}^{\text{ship_wake}}\in\xi_w^{\text{ship_wake}}(j=1,\ldots,l_2)$, and $\eta_{w_j}^{\text{ship_wake}}\in\eta_w^{\text{ship_wake}}(j=1,\ldots,l_2)$. S and W are the labels of the ship and ship wake after clustering, respectively.

Then, the second target flag is expressed as

$$F_2\left(\xi_{s_i}^{\text{ship}},\eta_{s_i}^{\text{ship}},n\right)=\begin{cases}1, & \text{if } \min\{l_{sw}\}<l_{\min}\\ 0, & \text{else}\end{cases}\tag{11}$$

and

$$F_2\left(\xi_{w_j}^{\text{ship_wake}},\eta_{w_j}^{\text{ship_wake}},n\right)=\begin{cases}-1, & \text{if } \min\{l_{sw}\}<l_{\min}\\ 0, & \text{else}\end{cases}\tag{12}$$

where $l_{sw}=((\xi_{s_i}^{\text{ship}}-\xi_{w_j}^{\text{ship_wake}})^2+(\eta_{s_i}^{\text{ship}}-\eta_{w_j}^{\text{ship_wake}})^2)^{1/2}$ and $l_{\min}$is a threshold factor. When $\min\{l_{st}\}<l_{\min}$, we redefine $\xi_{s_j}^{\text{wake}}=\xi_{w_j}^{\text{ship_wake}}$ and $\eta_{s_j}^{\text{wake}}=\eta_{w_j}^{\text{ship_wake}}$. $(\xi_{s_j}^{\text{wake}},\eta_{s_j}^{\text{wake}})$ and $(\xi_{s_i}^{\text{ship}},\eta_{s_i}^{\text{ship}})$ are the pixels of the ship wake and ship belonging to ship "S" in the raw observed TB maps.

By step 3, the ships in the raw observed TB maps can be accurately detected, and the pixels of the ship-self and ship wake are flagged as "1" or "−1." Meanwhile, the false flag that "ship-free" pixels falsely flagged as "1" in the first target flag is also reflagged as "0" by the second target flag to reduce the FAR.

D. Step 4: Estimating Ship Headings

In step 3, the pixels of the ship-self and ship wake of a sailing ship are flagged as "1" or "−1." In this step, the ship heading can be estimated based on the flagged pixels ("1" or "−1") associated with the sailing ship.

The centers of the ship-self and ship wake in the observed TB maps are estimated by

$$\left(\xi_s^{\text{ship}},\eta_s^{\text{ship}}\right)=\left(\frac{1}{l_1}\sum_{i=1}^{l_1}\xi_{s_i}^{\text{ship}},\frac{1}{l_1}\sum_{i=1}^{l_1}\eta_{s_i}^{\text{ship}}\right)\tag{13}$$

$$\left(\xi_s^{\text{wake}},\eta_s^{\text{wake}}\right)=\left(\frac{1}{l_2}\sum_{j=1}^{l_2}\xi_{s_j}^{\text{wake}},\frac{1}{l_2}\sum_{j=1}^{l_2}\eta_{s_j}^{\text{wake}}\right).\tag{14}$$

Then, the ship heading of ship "S" is estimated by

$$\vec{H}_s=\left(\xi_s^{\text{ship}}-\xi_s^{\text{wake}},\eta_s^{\text{ship}}-\eta_s^{\text{wake}}\right).\tag{15}$$

In this step, the ship heading can be obtained based on a single raw observed TB map.

E. Step 5: Estimating Ship Trajectory

The consecutive observed TB maps $(t_1,\ldots,t_s)$ that contain a ship are flagged in step 1, and the location of the ship is also determined in step 3. Thus, the ship trajectory can be estimated by the second target flag map of consecutive raw observed TB maps that contain ships using the "and" operation as follows:

$$T_{\text{ship}}(\xi,\eta)=\sum_{n=t_1}^{t_s}F_2(\xi,\eta,n).\tag{16}$$

Moreover, since the ship and the ship wake are flagged as "1" and "−1," respectively, the ship trajectory can be estimated based on the ship and the ship wake.

The ship trajectory based only on the ship is estimated by

$$T_{\text{ship_self}}(\xi,\eta)=\begin{cases}1, & \text{if } T_{\text{ship}}(\xi,\eta)>=1\\ 0, & \text{else}\end{cases}\tag{17}$$

and the ship's trajectory based only on the ship's wake is estimated by

$$T_{\text{ship_wake}}(\xi,\eta)=\begin{cases}1, & \text{if } T_{\text{ship}}(\xi,\eta)<=-1\\ 0, & \text{else.}\end{cases}\tag{18}$$

In the proposed ship detection algorithm, since the ship-containing pixels in the first target flag must satisfy the conditions in (11), some true "ship-containing" pixels may be falsely flagged as "0" in the second target flag, which will increase the missed detection probability. Thus, the scaling threshold factors $M1$ and $M2$ should be carefully chosen to reduce the FAR and missed detection probability at the same time. In addition, a nearest neighbor object association algorithm can be applied to the ship trajectory in (17) or (18) to reduce the missed detection probability [24].

In contrast to the algorithm in [17], the improved ship detection algorithm shows better performance in terms of the low FAR value, and the ship heading can be simply obtained based only on a single raw observed TB map.

IV. Experiments and Results

In this section, the improved ship detection algorithm based on ship wakes is verified by a series of airborne experiments, and its performance is assessed. In airborne experiments, an X-band PIMS is installed on the bottom of an aircraft.

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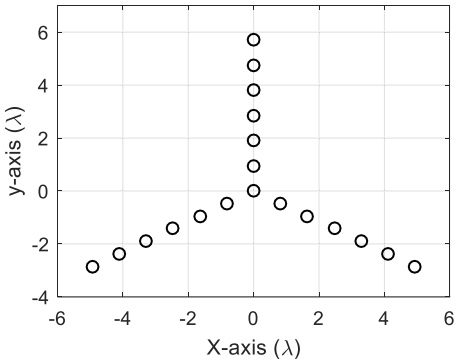

Fig. 3. Antenna array of X-band PIMS with each arm of six antennas.

Fig. 4. Aircraft used in the experiments.

The X-band PIMS employs a "Y-shaped" antenna array with each arm having six antennas, as shown in Fig. 3. The −3 dB beamwidth of each antenna is approximately 65°, and the center frequency and the bandwidth are 10.65 GHz and 100 MHz, respectively. In the airborne experiments, the integration time is 100 ms, which can be adjusted as needed. For comparison, an optical camera is installed on the bottom of the aircraft, as shown in Fig. 4.

In the airborne experiments, the external calibration method [25], the FFT [26], and the absolute calibration method [27] are performed to obtain the raw observed TB maps. Since the angular resolution of the X-band PIMS is poor at approximately 7.2° in the Blackman window [28], an FFT is performed with threefold zero-padding.

In this section, the performance of the improved ship detection algorithm is assessed in terms of the detection probability and FAR. The detection probability is defined as the ratio $(S_d/S \cdot 100\%)$ of the raw observed TB maps S_d with ships correctly flagged as "1" to the total raw observed TB maps of the ship S, and FAR is defined as the ratio $(S_f/S \cdot 100\%)$ of the raw observed TB maps S_f with "ship-free" pixels falsely flagged as "1" to the total observed TB maps of the ship S [29].

A. Experiment Results

In the airborne experiments, the improved ship detection algorithm is performed. For the raw observed TB maps in one airborne experiment, step 1 (searching targets) is first implemented with $N{\cdot}\sigma_0 = 1.0$ K, and several targets are found in the raw observed TB maps, the pixels are flagged as "1," and the maps are considered observed TB maps that contain ships. The frame ranges of the several identified targets flagged as "1" are shown in Table I. Four raw observed TB maps, frame 6580, frame 6595, frame 6610, and frame 6625, that represent Ship 5 in Table I are shown in Fig. 5(a), and it is found that residual systematic errors and systematic noise exist in the raw observed TB maps and that ships cannot be clearly seen in the raw observed maps [17], as expected. The difference-detection TB maps of the raw observed TB maps in Fig. 5(a) are shown in Fig. 5(b). Since $|n-m|$ mainly depends on the fluctuations of the aircraft and can be determined by analyzing the TB fluctuations of one element in the PIMSs, $|n-m|$ is set to 10 in this step, and, for a fair comparison, this is the same value used in [17]. Some pixels with "cold" TB characteristics or "hot" TB characteristics are found in the difference-detection TB maps, in which the maximum absolute TB values are higher than the threshold $N \cdot \sigma_0 = 1.0$ K in (8). Then, step 2 (improving TB maps) is performed, in which the "ship-free" observed TB map of frame 6300 is carefully selected as the sea background $TB_{map}^{\text{sea}}(\xi, \eta, k)$ in (9) by analyzing the TB fluctuations of the hub elements of the X-band PIMS (the same as in [17, Figs. 11 and 15]). As shown in Fig. 5(c), the improved observed TB maps of the four raw observed TB maps are given. The "cold" TB regions in the improved TB maps are always accompanied by the "hot" TB regions, as expected. In the improved observed TB maps, the "cold" TB regions may be caused by metal ships, and the "hot" TB regions may be caused by ship wakes or islands. Then, step 3 (clustering targets) is performed, in which the first target flag is obtained with $M1 \cdot \sigma_0 = M2 \cdot \sigma_0 = 1.3$ K, and the second target flag is obtained with $l_{\min} = 0.15$. As shown in Fig. 5(d), the second target flag of the four improved observed TB maps in Fig. 5(c) is given. After that, step 4 (estimating ship headings) is performed, and the estimated ship headings of the four improved observed TB maps in Fig. 5(c) are represented by the red arrows in Fig. 5(d). For comparison, the corresponding optical images of the raw observed TB maps [see Fig. 5(a)] are also given in Fig. 5(e). It can be seen that there is indeed a metallic ship on the sea in the four optical images, in which the location of the ship and its wake are consistent with the flag of the ship and its wake in the second target flag map [see Fig. 5(d)]. In addition, the estimated ship heading in each improved observed TB map [see Fig. 5(d)] is consistent with the corresponding optical image [see Fig. 5(f)]. After the improved ship detection algorithm applied to the residual raw observed TB maps of Ship 5 in Table I, the ship trajectory is estimated, as shown in Fig. 6(a), in which the black stars denote the mean of the "ship-containing" pixels belonging to a ship. The second target flag and estimated ship heading of the residual raw observed TB maps of Ship 5 in

TABLE I
FRAME RANGE OF SEVERAL CONSECUTIVE TB MAP SETS FLAGGED AS "1" IN ONE AIRBORNE EXPERIMENT

No.	Frame range
Ship 1	1500~1650
Ship 2	2300~2400
Ship 3	3300~3400
Ship 4	4040~4140
Ship 5	6500~6650

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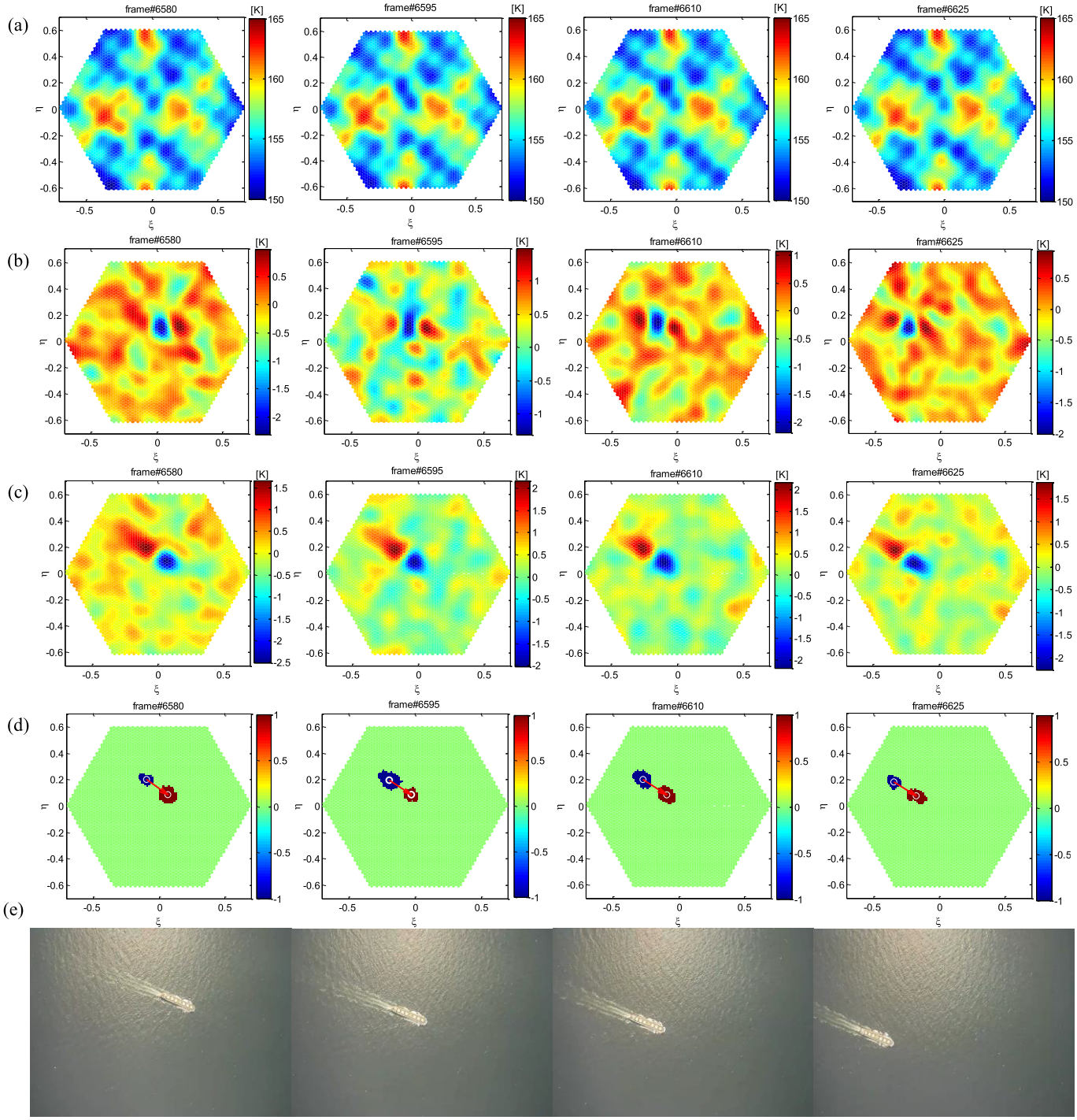

Fig. 5. (a) Four originated TB maps of a consecutive TB map set flagged as "1" in step 1 in the airborne experiments. (b) Adjacent difference TB maps of the four examples by step 1 of the proposed ship detection algorithm with n–m = 10. (c) Background cancellation TB maps of the four examples by step 2 of the proposed ship detection algorithm. (d) Target flag maps of the four examples by step 3 of the proposed ship detection algorithm. (e) Optical images of the four examples.

Table I are demonstrated by the corresponding optical images. The experiment results indicate that ships can be accurately detected by the improved ship detection algorithm and that the ship heading can also be accurately estimated using a single snapshot obtained by the improved ship detection algorithm.

The ship trajectory maps of the other three ships obtained by the improved ship detection algorithm are also shown in Fig. 6(a). The ship heading of four ships in Fig. 6(a) obtained by the improved ship detection algorithm is shown in Fig. 6(b), in which the radius of the polar figures denotes the absolute value of the estimated ship heading ($\vec{H}_s$ denotes the direction cosines). For comparison, the optical images of the ships within the middle of the FOV are given in Fig. 6(c). The ship heading obtained by the proposed algorithm is consistent with the ship heading in the optical images. The standard deviations of the estimated ship heading of the four ships are

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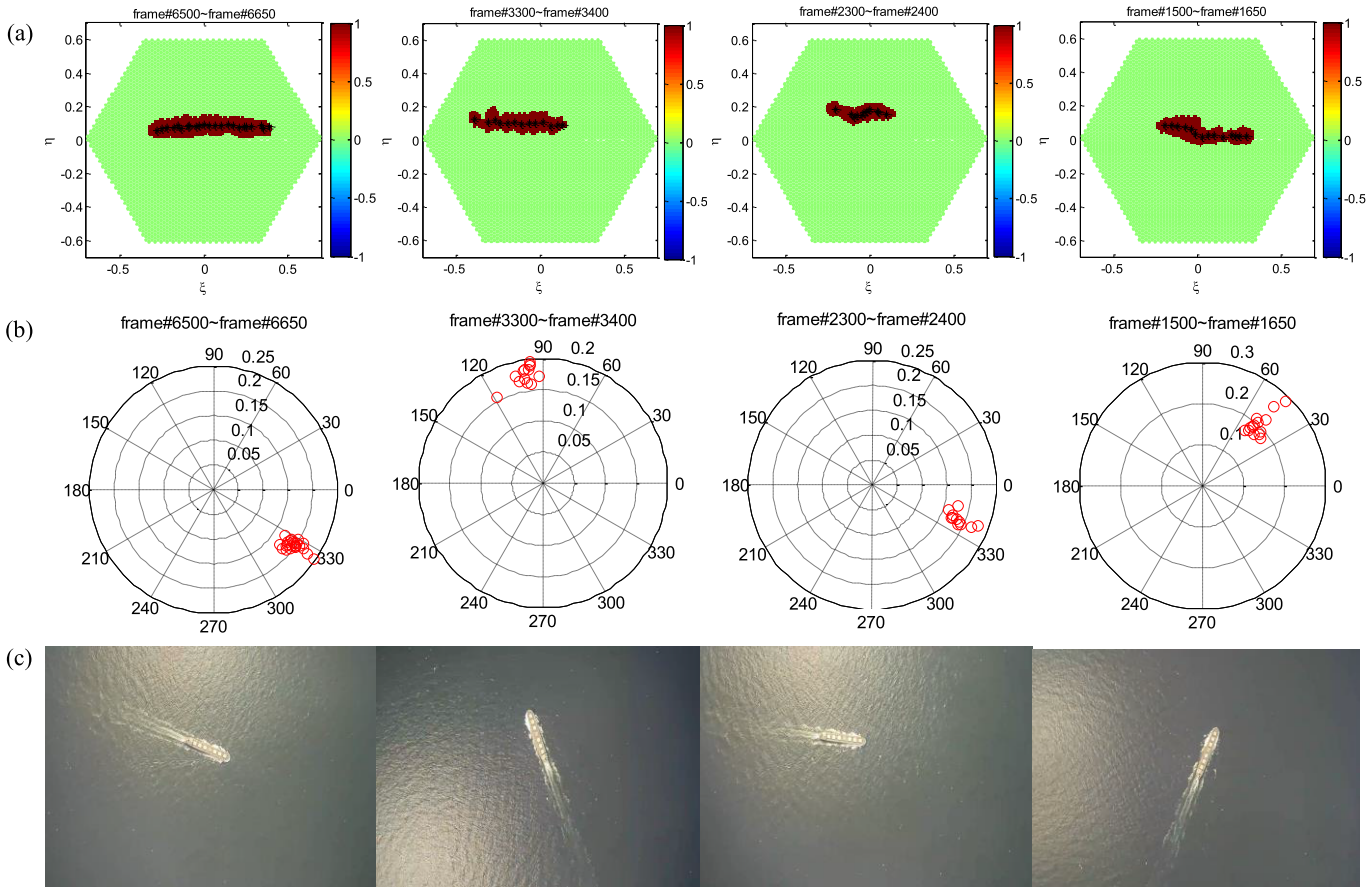

Fig. 6. (a) Ship tracking maps, (b) estimated ship heading, and (c) optical images of four ships in Table I detected by the ship detection algorithm.

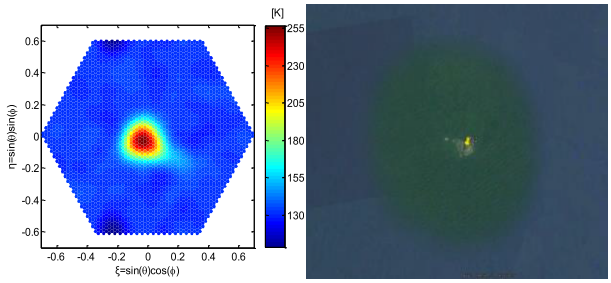

Fig. 7. (Left) Raw observed TB map and (right) optical image of an island in one airborne experiment.

approximately 2.67°, 6.24°, 3.03°, and 3.96°. All experiment results indicate that the proposed ship detection algorithm is feasible and can accurately estimate the ship heading and ship trajectory.

In addition, the islands in the sea exhibit a "hot" TB characteristic in the observed TB maps, which is similar to the TB characteristic of ship wakes. As shown in Fig. 7, the raw observed TB map and the optical map of an island are given. In the improved ship detection algorithm, islands can also be detected but are not falsely flagged as ships, which is different from ship detection by SAR.

B. Impacts of the Algorithm Parameters on the Detection Performance

As mentioned above, the algorithm parameters (N, $M1$, and $M2$) have nonnegligible impacts on the detection performance. The impact of the parameter N was analyzed in [17], and it has the same impact on the improved ship detection algorithm. Thus, the residual algorithm parameters ($M1$ and $M2$) are analyzed in this section. As shown in Fig. 8 (top), the ship trajectory maps of Ship 5 when $M_2 \cdot \sigma_0 = 0.8$ K and $l_{\min} = 0.15$ with different algorithm parameters $M_1 \cdot \sigma_0 = 0.9$, 0.7, 0.5, and 0.3 K are shown. As shown in Fig. 8 (bottom), the ship trajectory maps of Ship 5 with $M_1 \cdot \sigma_0 = 0.6$ K and $l_{\min} = 0.15$ and different algorithm parameters $M_2 \cdot \sigma_0 = 0.9$, 0.7, 0.5, and 0.3 K are provided. It can be seen that the ship trajectory maps exhibit different characteristics as the algorithm parameters $M_1 \cdot \sigma_0$ and $M_2 \cdot \sigma_0$ change. The detection probability and FAR of Ship 5 in Fig. 8 versus the different values of $M_2 \cdot \sigma_0$ are given in Tables II and III. It is found that the lower the values of $M1$ and $M2$ are, the higher the FAR and the higher the detection probability, as expected. Thus, the values of $M1$ and $M2$ should be carefully considered to obtain a high ship detection probability with a low ship FAR value in the proposed algorithm.

In addition, it must be noted that aircraft fluctuations have a negative impact on the improved ship detection algorithm, which is similar to the ship detection algorithm in [17]. Thus, the sea background must be carefully selected in step 2 to obtain a high contrast in the improved observed TB map by analyzing the TB fluctuations of one element in the PIMSs, which is also the same as the previous detection algorithm, and this was analyzed and discussed in detail in [17].

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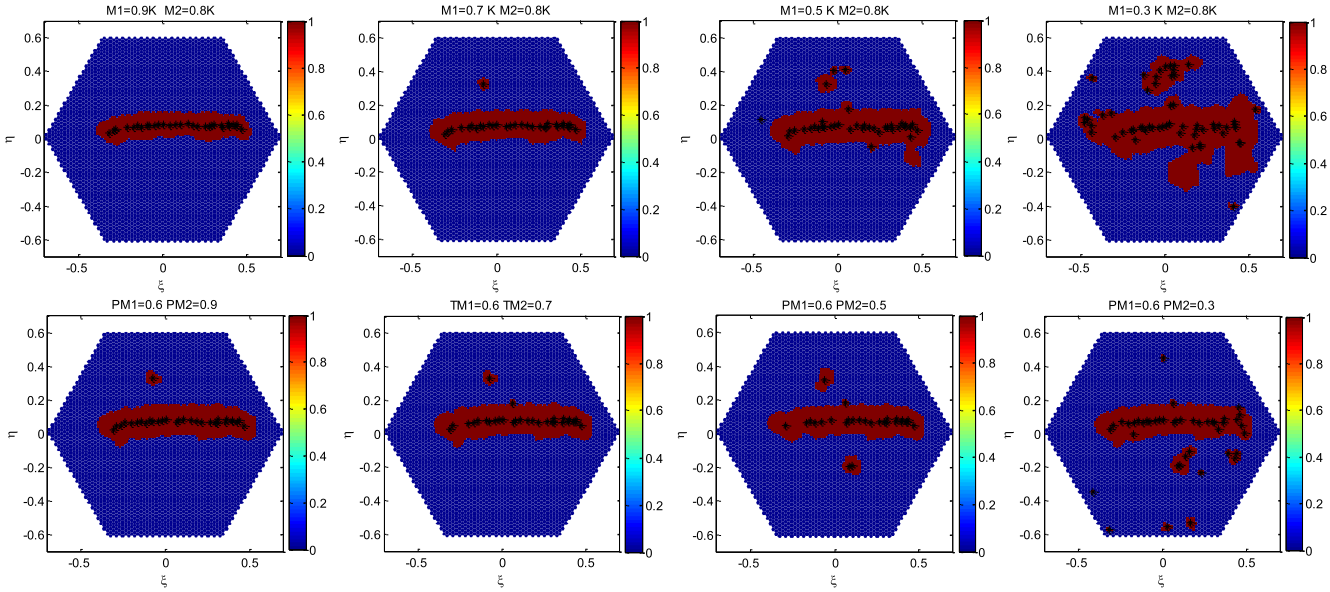

Fig. 8. (Top) Ship trajectory maps of ship 5 with the same algorithm M_2 $\sigma_0 = 0.8$ K and different algorithm parameters M_1 $\sigma_0 = 0.9$, 0.7, 0.5, and 0.3 K. (Bottom) Ship trajectory maps of ship 5 with the same algorithm M_1 $\sigma_0 = 0.6$ K and different algorithm parameters M_2 $\sigma_0 = 0.9$, 0.7, 0.5, and 0.3 K.

TABLE II

DETECTION PROBABILITY AND FAR OF SHIP 5 IN FIG. 8 (TOP) WITH DIFFERENT PARAMETERS $M_1 \cdot \sigma_0$

$M_1 \cdot \sigma_0$ (K)	M_2*σ_0=0.8 K	
	Detection Probability (%)	FAR (%)
0.9	96.8	0
0.7	96.8	3.2
0.5	96.8	22.6
0.3	100.0	93.5

TABLE III

DETECTION PROBABILITY AND FAR OF SHIP 5 IN FIG. 8 (BOTTOM) WITH DIFFERENT PARAMETERS $M_2 \cdot \sigma_0$

$M_2 \cdot \sigma_0$ (K)	M_1*σ_0=0.6 K	
	Detection Probability (%)	FAR (%)
0.9	96.7	3.2
0.7	96.7	6.5
0.5	96.7	9.7
0.3	100	41.9

C. Performance of the Improved Ship Detection Algorithm

As mentioned above, the improved ship detection algorithm is expected to have better performance than the algorithm in [17]. Thus, the performance of the improved ship detection algorithm is analyzed and compared to the performance of the only existing ship detection algorithm in [17].

As mentioned in Section III, the second target flag in step 3 (clustering targets) can reduce the FAR. As shown in Fig. 9 (top), the ship flag maps of four examples (frame 2295, frame 2300, frame 2315, and frame 2335) detected by the algorithm in [17] are given for $N \cdot \sigma_0 = 1.0$ K and $M \cdot \sigma_0 = 0.6$ K. Moreover, the second target flag of four examples (frame 2295, frame 2300, frame 2315, and frame 2335) detected by the improved ship detection algorithm are given in Fig. 9 (bottom) for $N \cdot \sigma_0 = 1.0$ K, $M_1 \cdot \sigma_0 = M_2 \cdot \sigma_0 = 0.6$ K, and $l_{\min} = 0.15$. The corresponding optical images indicate that there is only a ship in the raw observed TB maps, and the ship is correctly detected by the two algorithms. However, some "ship-free" pixels in the ship flag maps of the algorithm in [17] are falsely flagged as "ship-containing" pixels, but no "ship-free" pixels in the second target flag of the improved algorithm are falsely flagged as "ship-containing" pixels or "ship wake-containing" pixels. As a result, the improved ship detection algorithm will obtain a lower FAR than the only existing ship detection algorithm [17].

The previous ship detection algorithm in [17] and the improved ship detection algorithm are applied to the raw observed TB maps of Ship 2 in Table I with different scaling threshold factors $M \cdot \sigma$ or $M_1 \cdot \sigma$. The ship trajectory maps of both ship detection algorithms are shown in Fig. 10 with $M \cdot \sigma$ (top) or $M_1 \cdot \sigma$ (bottom) = 1.5, 1.2, 0.9, and 0.6 K at $M_2 \cdot \sigma_0 = 0.8$ K. It can be seen that more "ship-free" pixels in the ship trajectory maps of the ship detection algorithm in [17] are falsely flagged as "ship-containing" pixels, which increases the ship FAR value, especially when the algorithm parameter values are low, i.e., $M \cdot \sigma = 0.9$ and 0.6 K. However, it can also be seen that few "ship-free" pixels in the ship trajectory maps of the improved ship detection algorithm are falsely flagged as "ship-containing" pixels or "ship wake-containing" pixels. Furthermore, the detection probability and FAR of Ship 2 in Fig. 10 are counted and shown in Table IV. As shown in Fig. 10 and Table IV, the experiment results indicate that the improved detection algorithm shows a better detection performance with a lower FAR value than the only existing ship detection algorithm in [17].

Moreover, the statistics of the ship detection probability and ship FAR of both ship detection algorithms are obtained with different algorithm parameters $M \cdot \sigma$ or $M1 \cdot \sigma$ for the

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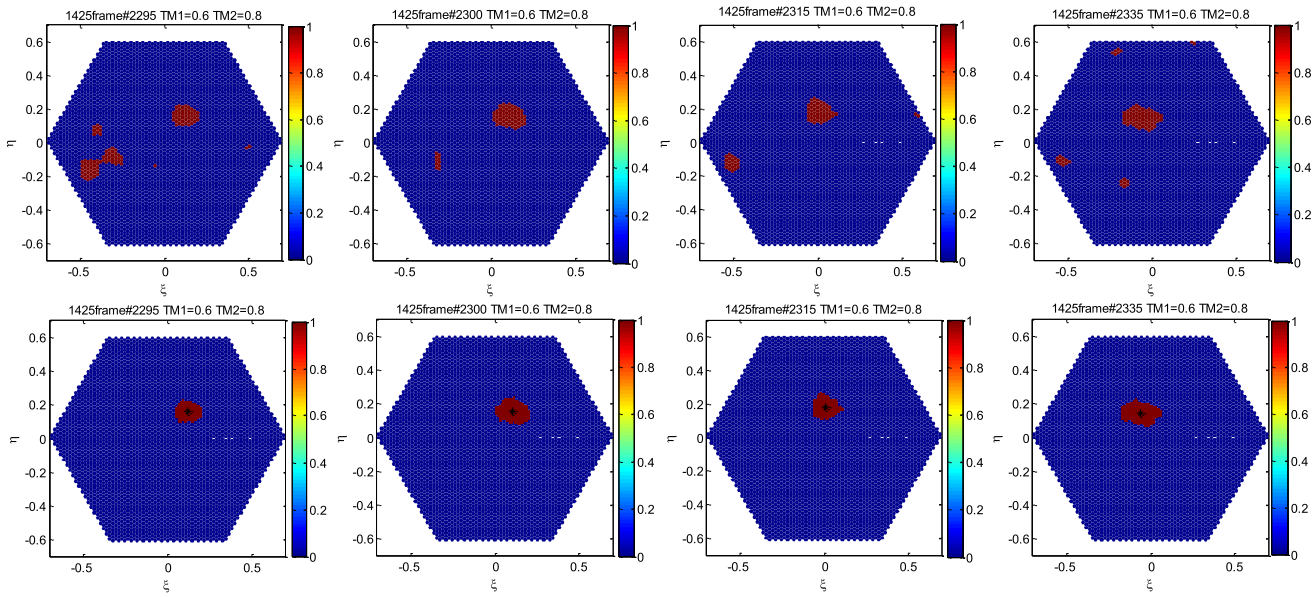

Fig. 9. Four examples (frame 2295, frame 2300, frame 2315, and frame 2335) of (top) ship flag maps using the previous algorithm with $N\ \sigma_0 = 1.0$ K and $M\ \sigma_0 = 0.6$ K, and (bottom) second target flag using the proposed algorithm with $N\ \sigma_0 = 1.0$ K, $M_1\ \sigma_0 = M_2\ \sigma_0 = 0.6$ K, and $l_{\min} = 0.15$.

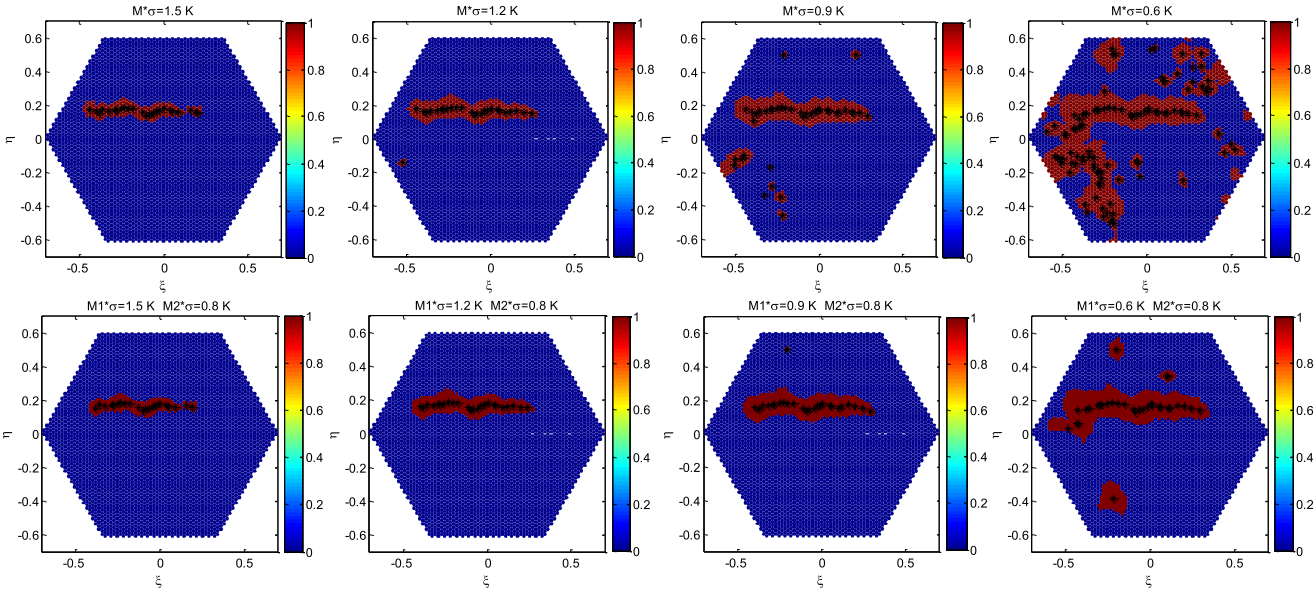

Fig. 10. (Top) Ship tracking maps of Ship 2 using the ship detection algorithm in [12] with $M\ \sigma_0 = 1.5$, 1.2, 0.9, and 0.6 K (from left to right), and (bottom) the ship tracking maps of Ship 2 using the improved ship detection algorithm with $M_1\ \sigma_0 = 1.5$, 1.2, 0.9, and 0.6 K (from left to right).

TABLE IV

DETECTION PROBABILITY AND FAR OF SHIP 2 IN TABLE I DETECTED BY THE ONLY EXISTING SHIP DETECTION ALGORITHM IN [17] AND THE IMPROVED SHIP DETECTION ALGORITHM

$M_1 \cdot \sigma_0$ (K)	Only existing ship detection algorithm in [17]		Improved ship detection algorithm	
	Detection Probability (%)	FAR (%)	Detection Probability (%)	FAR (%)
1.5	88.9	0	77.8	0
1.2	96.3	3.7	88.9	0
0.9	100.0	33.3	92.6	3.7
0.6	100.0	96.3	100.0	22.2

airborne experiments. As shown in Fig. 11, the ship detection probability (top) and ship FAR (bottom) of both ship detection algorithms as a function of the algorithm parameter $M \cdot \sigma$ or $M1 \cdot \sigma$ ($\sigma = 0.65$ K) are given for Ship 2 (left) and Ship 5 (right) in Table I, respectively. It is found that the ship detection probability and ship FAR in both ship detection algorithms decrease as the algorithm parameter $M \cdot \sigma$ or $M1 \cdot \sigma$ increases, as expected. Therefore, the algorithm parameter $M1$ should also be carefully considered to obtain a high ship detection probability with a low ship FAR in the improved ship detection algorithm; for example, $M \cdot \sigma = 1.0$ K in Fig. 11.

In addition, it can be seen that the detection probability of the proposed improved algorithm is slightly lower than the detection probability of the previous algorithm in [17]. However, it can also be seen that the ship FAR of the improved ship detection algorithm is much lower than the ship FAR of the ship detection algorithm in [17] with the same

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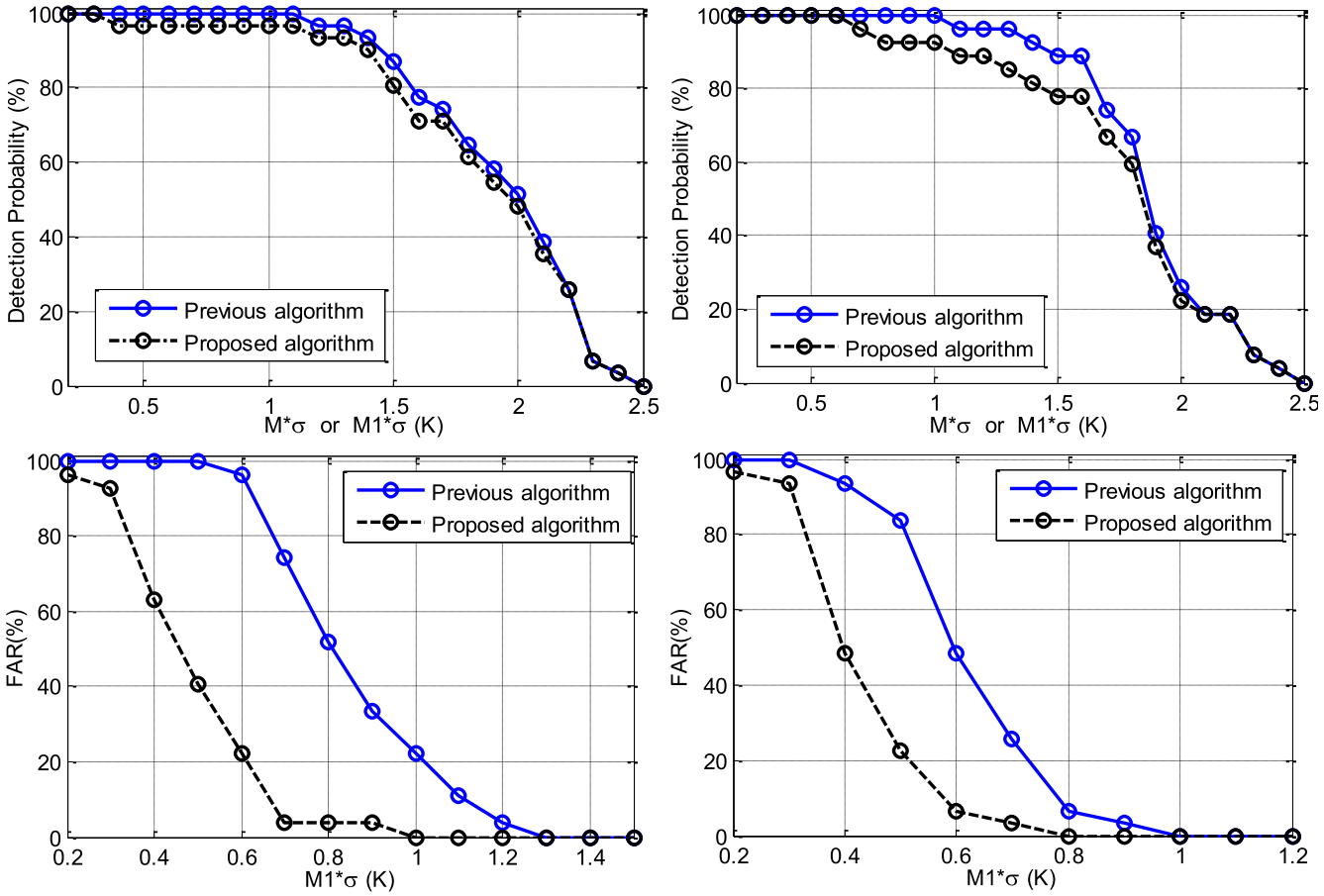

Fig. 11. (Top) Ship detection probability and (bottom) ship FAR of the two examples of (left) Ship 2 and (right) Ship 5 in Table I as a function of the algorithm parameter M σ_0 or M_1 σ_0 using the previous algorithm and the proposed algorithm.

algorithm parameters. The experiment results demonstrate that the improved ship detection algorithm has a better performance with a lower FAR value than the only existing ship detection algorithm in [17], as expected.

V. Conclusion

Airborne PIMSs were proposed as a powerful complementary tool for ship detection, and the only existing ship detection algorithm was proposed by Lu et al. [17] for ship detection by airborne PIMSs, which is only based on the "cold" TB characteristic of metal ships in PIMIs. Moreover, it has been found that ship wakes always exhibit a "hot" TB characteristic in PIMIs. To further improve the detection performance for ship detection by airborne PIMSs, an improved ship detection algorithm is proposed, which is simultaneously based on the "cold" TB characteristic of ships and the "hot" TB characteristic of ship wakes in PIMIs. A microwave TB model of metal ships with wakes is established to analyze the TB characteristics of metal ships and ship wakes. Based on the model and characteristics of metal ships with ship wakes, an improved ship detection algorithm is proposed, which consists of five steps: searching targets, improving TB maps, clustering targets, estimating ship headings, and estimating ship trajectories. In contrast to the only existing ship detection algorithm in [17], the improved algorithm is expected to have better performance with a lower FAR value, and the ship heading can be accurately estimated using only a single snapshot by the improved algorithm. The improved algorithm is verified by airborne experiments using an X-band PIMS. The experiment results demonstrate that ships can be accurately detected, and ship trajectories can be accurately estimated by the improved algorithm. The experiment results indicate that the ship headings can be accurately estimated using a single snapshot by the improved algorithm. The impact of the algorithm parameters ($M1$ and $M2$) on the detection performance is analyzed and discussed in terms of the detection probability and FAR. The analytical results indicate that the algorithm parameters $M1$ and $M2$ should be carefully selected to obtain a high ship detection probability with a low ship FAR in the proposed algorithm. Furthermore, the detection performance of the improved algorithm is analyzed and compared with the only existing ship detection algorithm in [17]. The experiment results indicate that the improved ship detection algorithm has a better performance with a lower FAR value than the only existing ship detection algorithm in [17], as expected.

References

[1] Z. Liu, H. Wang, L. Weng, and Y. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.

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[2] F. Yang, Q. Xu, and B. Li, "Ship detection from optical satellite images based on saliency segmentation and structure-LBP feature," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 602–606, May 2017.

[3] G. K. Høye, T. Eriksen, B. J. Meland, and B. T. Narheim, "Space-based AIS for global maritime traffic monitoring," *Acta Astronautica*, vol. 62, nos. 2–3, pp. 240–245, Jan. 2008.

[4] M. D. Graziano, M. D'Errico, and E. Razzano, "Constellation analysis of an integrated AIS/remote sensing spaceborne system for ship detection," *Adv. Space Res.*, vol. 50, no. 3, pp. 351–362, Aug. 2012.

[5] C. Zhu, H. Zhou, R. Wang, and J. Guo, "A novel hierarchical method of ship detection from spaceborne optical image based on shape and texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3446–3456, Sep. 2010.

[6] J.-W. Lu, Y.-J. He, H.-Y. Li, and F.-L. Lu, "Detecting small target of ship at sea by infrared image," in *Proc. IEEE Int. Conf. Autom. Sci. Eng.*, Shanghai, China, Mar. 2006, pp. 165–169.

[7] H. Wang, Z. Zou, Z. Shi, and B. Li, "Detecting ship targets in spaceborne infrared image based on modeling radiation anomalies," *Infr. Phys. Technol.*, vol. 85, pp. 141–146, Sep. 2017.

[8] S. Brusch, S. Lehner, T. Fritz, M. Soccorsi, A. Soloviev, and B. van Schie, "Ship surveillance with TerraSAR-X," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 3, pp. 1092–1103, Mar. 2011.

[9] Y. Wang and H. Liu, "A hierarchical ship detection scheme for high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 4173–4184, Oct. 2012.

[10] L. Yujiri, M. Shoucri, and P. Moffa, "Passive millimeter-wave imaging," *IEEE Micro.*, vol. 4, no. 3, pp. 39–50, Sep. 2003.

[11] Y. A. Pirogov, "Passive millimeter-wave imaging," *Radiophys. Quantum Electron.*, vol. 46, nos. 8–9, pp. 594–603, Aug./Sep. 2003.

[12] A. Di Simone, H. Park, D. Riccio, and A. Camps, "Sea target detection using spaceborne GNSS-R delay-Doppler maps: Theory and experimental proof of concept using TDS-1 data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 9, pp. 4237–4255, Sep. 2017.

[13] C. S. Ruf, C. T. Swift, A. B. Tanner, and D. M. Le Vine, "Interferometric synthetic aperture microwave radiometry for the remote sensing of the Earth," *IEEE Trans. Geosci. Remote Sens.*, vol. GRS-26, no. 5, pp. 597–611, Sep. 1988.

[14] R. Appleby, D. G. Gleed, R. N. Anderton, and A. H. Lettington, "High-performance passive millimeter-wave imaging," *Opt. Eng.*, vol. 32, no. 6, pp. 1370–1373, May 1993.

[15] A. R. Harvey and R. Appleby, "Passive mm-wave imaging from UAVs using aperture synthesis," *Aeronaut. J.*, vol. 107, no. 1068, pp. 87–98, Feb. 2003.

[16] H. Lu et al., "Ship detection by an airborne passive interferometric microwave sensor (PIMS)," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2682–2694, Apr. 2020.

[17] H. Lu, Y. Li, and H. Li, "A ship detection and detection algorithm for an airborne passive interferometric microwave sensor (PIMI)," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3519–3532, 2021.

[18] N. A. Salmon, "Outdoor passive millimeter-wave imaging: Phenomenology and scene simulation," *IEEE Trans. Antennas Propag.*, vol. 66, no. 2, pp. 897–908, Feb. 2018.

[19] J. Gourrion, R. Sabia, M. Portabella, J. Tenerelli, S. Guimbard, and A. Camps, "Characterization of the SMOS instrumental error pattern correction over the ocean," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 4, pp. 793–797, Jul. 2012.

[20] Y. Sasaki, I. Asanuma, K. Muneyama, G. Naito, and T. Suzuki, "The dependence of sea-surface microwave emission on wind speed, frequency, incidence angle, and polarzation over the frequency range from 1 to 40 GHz," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-25, no. 2, pp. 138–146, Mar. 1987.

[21] J. Gourrion, S. Guimbard, M. Portabella, and R. Sabia, "Toward an optimal estimation of the SMOS antenna-frame systematic errors," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4752–4760, Sep. 2013.

[22] P. A. Hwang, "Surface foam and L-band microwave radiometer measurements in high winds," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2766–2776, May 2019.

[23] H. Lang, Y. Xi, and X. Zhang, "Ship detection in high-resolution SAR images by clustering spatially enhanced pixel descriptor," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5407–5423, Aug. 2019.

[24] K. Zhang, H. Lin, G. Liu, and X. Ma, "Improved interacting multiple model-new nearest neighbor data association algorithm," in *Proc. IEEE 4th Int. Conf. Control Sci. Syst. Eng. (ICCSSE)*, Aug. 2018, pp. 499–504.

[25] H. Lu et al., "A hybrid calibration method for aperture synthesis radiometers," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 651–655, May 2016.

[26] E. Anterrieu and A. Khazaal, "Brightness temperature map reconstruction from dual-polarimetric visibilities in synthetic aperture imaging radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 3, pp. 606–612, Mar. 2008.

[27] M. A. Brown, F. Torres, I. Corbella, and A. Colliander, "SMOS calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 3, pp. 646–658, Mar. 2008.

[28] A. Camps, I. Corbella, J. Bara, and F. Torres, "Radiometric sensitivity computation in aperture synthesis interferometric radiometry," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 2, pp. 680–685, Mar. 1998.

[29] S. Zhao, Z. Zhang, W. Guo, and Y. Luo, "An automatic ship detection method adapting to different satellites SAR images with feature alignment and compensation loss," *IEEE Trans. Geosci. Remote Sens.*, vol. 60 2022, Art. no. 5225217.

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